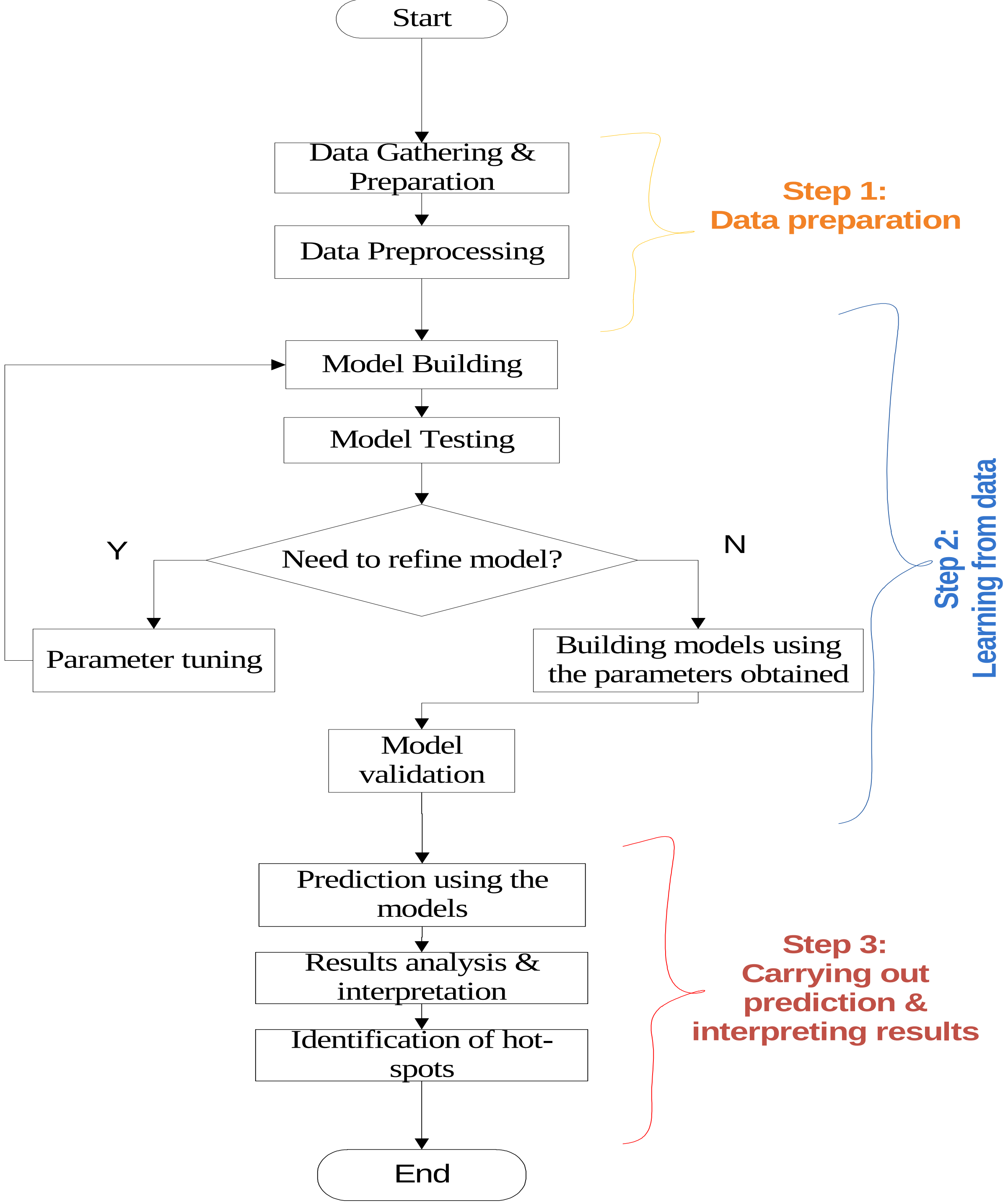
CS 522 – Selected Topics in CS

Lecture 04 – Supervised Learning

February 6, 2022

Generic Machine Learning Framework



+ Supervised Learning

- **Given:** Training Examples
  - $\{(x_1, f(x_1)), (x_2, f(x_2)), \ldots\ldots, (x_p, f(x_p))\}$
    - For some unknown function (system) ***y=f(x)***
- **Find f(x)?**
  - Predict ***ý = f(x`),*** where x` is not in the training set.
- Supervised Learning is ***used*** for
  - Prediction of future cases
  - Knowledge Extraction

+ Supervised Learning

- Given: Examples ***(x, f(x))*** of some unknown function f
- Find: A good ***approximation*** of f, ***x*** provides ***some representation*** of the ***input***
  - The process of ***mapping*** a ***domain element*** into a ***representation*** is called ***Feature Extraction.***
  - $x \in \{0,1\}_n$ or $x \in R_n$ or $x \in$ categorical data
- ***The target function (label)***
  - $f(x) \in \{-1,+1\}$ Binary Classification
  - $f(x) \in \{1,2,3,..,k\}$ Multi-class classification
  - $f(x) \in R$ Regression

+ Types of supervised Learning

- Classification: $\mathbf{Y} \in \{1, ..., C\}$
- Regression: $\boldsymbol{Y} \in \Re$
- ***Example:***
  - If I want to predict whether ***a patient will die*** from a disease within six months, that is ***classification***
  - If I want to predict ***how long the patient will live***, that is ***regression***

+ Activity-1 (Examples for Classification & Regression)

- ***Disease diagnosis***
  - x: Properties of patient (symptoms, lab tests)
  - f : Disease (or maybe: recommended therapy)
- ***Part–of–Speech tagging***
  - x: An English sentence (e.g., The can will rust)
  - f : The part of speech of a word in the sentence
- ***Face recognition***
  - x: Bitmap picture of person’s face
  - f : Name the person (or maybe: a property of)
- ***Automatic Steering***
  - x: Bitmap picture of road surface in front of car
  - f : Degrees to turn the steering wheel

+ Representation

- ***Definition of thing or things to be predicted***
  - Classification: *classes*
  - Regression: *regression variable*
- ***Definition of things (instances) to make predictions for***
  - Individuals
  - Families
  - Neighborhoods, etc.
- ***Choice of descriptors (features) to describe different aspects of instances***

+ Formal description

- Defining ***X*** as a set of ***instances, x*** described by ***features***
- Given training examples ***D*** from ***X***
- Given a ***target function c*** that maps ***X–>{0,1}***
- Given a ***hypothesis space H***
- Determine a hypothesis ***h*** in ***H*** such that ***h(x)=c(x)*** for all ***x*** in ***D.***
- ***Hypothesis Space***
  - The ***hypothesis space*** determines the ***functional*** form
  - It defines what are ***allowable rules/functions*** for classification
  - ***Each classification*** method uses a ***different hypothesis space***

+ Inductive learning

- Data produced by ***"target".***
- Hypothesis learned from data in order to ***"explain", "predict", "model" or "control"*** target.
- ***Generalisation*** ability is ***essential.***
- Inductive learning hypothesis:

"If the hypothesis works for enough data then it will work on new examples."

- It could also be stated thus:

"Any hypothesis found to approximate the target function well over a sufficiently large set of training examples will also approximate the target function over other unobserved examples"

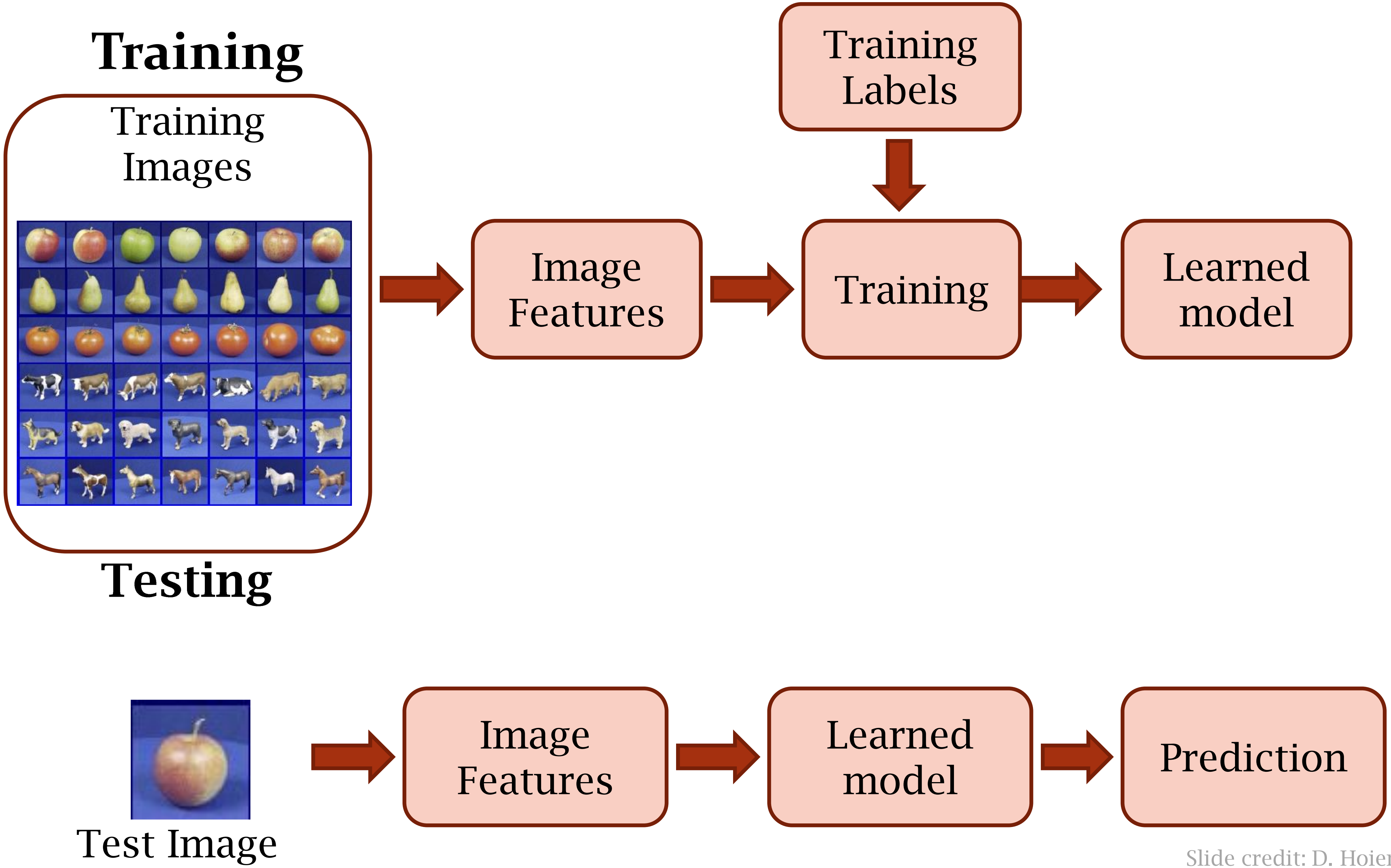
+ Inductive learning Cont'd…

- Given the following
  - $f{:}x \rightarrow y$
- If ***y*** is ***discrete/categorical*** variable, then this is ***classification*** problem.
- If ***y*** is ***real number/continuous***, then this is a ***regression*** problem.

Classification

- *How would you write a program to distinguish a picture of me from a picture of someone else?*
  - Provide examples pictures of me and pictures of other people and let a classifier learn to distinguish the two.
- *How would you write a program to determine whether a sentence is grammatical or not?*
  - Provide examples of grammatical and ungrammatical sentences and let a classifier learn to distinguish the two.
- *How would you write a program to distinguish cancerous cells from normal cells?*
  - Provide examples of cancerous and normal cells and let a classifier learn to distinguish the two.

+ Steps in Supervised Learning



Slide credit: D. Hoiem and L. Lazebnik

+ Examples (Weather Prediction)

Review

Three principal components:

1. Class label (aka "***label***", denoted y)
2. Features (aka "***attributes***")
3. Feature values (aka "***attribute values***", denoted x)

Features can be ***binary***, ***nominal*** or ***continuous***

A ***labeled dataset*** is a collection of (x; y) pairs

Class	Outlook	Temperature	Windy?
Play	Sunny	Low	Yes
No Play	Sunny	High	Yes
No Play	Sunny	High	No
Play	Overcast	Low	Yes
Play	Overcast	High	No
Play	Overcast	Low	No
No Play	Rainy	Low	Yes
Play	Rainy	Low	No

Examples (Weather Prediction)

Class	Outlook	Temperature	Windy?
Play	Sunny	Low	Yes
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Play	Rainy	Low	No

Review

Task: Predict the class for this "test" example.

Class	Outlook	Temperature	Windy?
???	Sunny	Low	No

Requires us to generalize from the training data

+ Classification

- **Whole idea**: ***Inject your knowledge into a learning system***

- **Sources of knowledge:**

1. Feature representation
- Not typically a focus of machine learning
- Typically seen as "***problem specific***"
- However, it's hard to learn on bad representations

2. Training data: labeled examples
- Often expensive to label lots of data
- Sometimes data is available for "free"

3. Model
- No single learning algorithm is always good ("***no free lunch***")
- Different learning algorithms work with different ways of representing the learned classifier
- When the data has nothing to say, which model is better
- Typically requires some control over ***generalization***

+ Classification – Supervised Learning

- Classification is the type of ***supervised Learning*** in which ***labelled data*** can ***use*** to make ***predictions*** in a ***non-continuous form***.

- In the classification technique, the algorithm ***learns*** from the ***data input*** given to it and then ***uses*** this ***learning*** to classify ***new observation***.

- This data set may merely be ***bi-class, or it may be multi-class*** too.

- ***Example:*** One of the examples of classification problems is to check whether the email is spam or not spam by train the algorithm for different spam words or emails.

+ Classification vs Regression

- Some algorithms can be used for ***both classification and regression*** with small modifications, such as ***decision trees and artificial neural networks***.
- Some algorithms cannot, or cannot easily be used for both problem types, such as ***linear regression*** for regression predictive modeling and ***logistic regression*** for classification predictive modeling.

Classification	Regression
Classification is the task of predicting a discrete class label.	Regression is the task of predicting a continuous quantity.
A classification algorithm may ***predict a continuous value***, but the continuous value is in the form of a ***probability for a class label***.	A regression algorithm may predict a ***discrete value***, but the discrete value in the form of an ***integer quantity***.
Classification predictions can be ***evaluated using accuracy***, whereas regression predictions cannot.	Regression predictions can be ***evaluated*** using ***root mean squared error,*** whereas classification predictions cannot.

+ Generalization

Training set (labels known)

Test set (labels unknown)

- How well does a learned model generalize from the data it was trained on to a new test set?

+ Remember…

- ***No classifier*** is inherently ***better*** than any other: you need to make assumptions to ***generalize***

- Three kinds of error
  - ***Inherent: unavoidable***
  - ***Bias: due to over-simplifications***
  - ***Variance: due to inability to perfectly estimate parameters from limited data***

+ How do we know if a model is performing well?

- A machine learning model's performance is ***considered good based*** on it's ***prediction*** and how well it ***generalizes*** on an ***independent*** test dataset.

- Based on the ***performance*** of ***different models*** we ***choose*** the model which ***ranks highest in performance***.

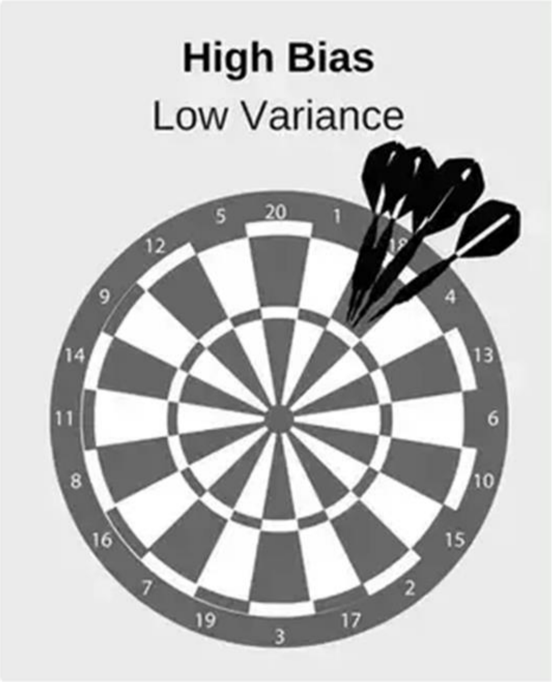
- *Example: Predicting the Election result*
  - Gather the data of one region and then train the system on the basis of single region data.
  - ***Can we apply this prediction to the entire county, state and then at national level?***
  - Sometimes results in the prediction error.

+ How do we know if a model is performing well?

- Assume an ***input x*** and we apply a ***function f*** on the input x to ***predict*** an output ***y***.
- ***Difference*** between the ***actual output*** and ***predicted output*** is the ***error***.
- Our ***goal*** with ***machine learning*** algorithm is to ***generate*** a ***model*** which ***minimizes the error*** of the test dataset.
  - ***Error*** = Σ(Actual output-predicted output)
  - ***Error*** = Reducible Error + Irreducible Error
    - ***Irreducible Error*** = Bias^2 + Variance

+ 1. Bias (Reducible Error)

- ***Average Predicted values are far off from actual values.***
- ***Model*** is too ***simple*** and ***doesn't capture*** the ***complexity*** results in underfitting
- The ***amount*** by which the ***expected model*** prediction differs from the ***actual value***.
- ***Model with high bias pay little attention to the data and oversimplifies the model***
- ***It always leads to high error on training and test data.***

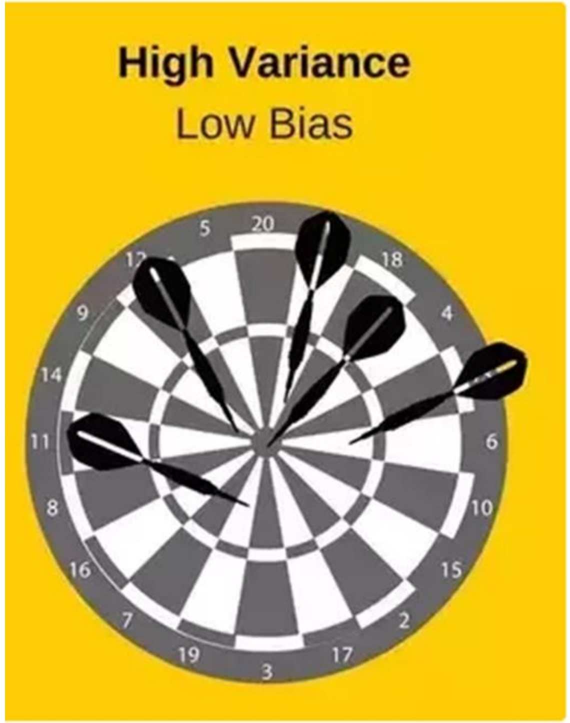


+ 2. Variance (Reducible Error)

- The ***amount*** by which the model ***prediction would change*** if we estimated it using ***different training dataset***.

- ***Model with high variance pays a lot of attention to training data and doesn't generalize on the data which it hasn't seen before***

- As a result, ***such models perform very well on training data but has high error rates on test data.***



+ Bias and Variance

- ***Overfitting*** : **Good** performance on the **training data** and **poor** results on **test data**.
- ***Underfitting:*** **Poor performance** on the **training** data and **poor result** on **test data**.
- ***Bias causes underfitting***
- ***Variance causes overfitting***

Low Bias-Low Variance

- This would have to be the **ideal situation**.
- The **error** of **prediction should** be as less as possible.
- Models are **accurate** and **consistent** on averages.
- **We strive for this in our model but it is not possible.**

+ Low Bias- High Variance

- ***Overfitting*** happens when our model captures the noise along with the underlying patterns in data.
- It happens when we train our model a lot over noisy dataset.
- Models are somewhat accurate but inconsistent on averages.
- A small change in the data can cause a large error.

High Bias – Low Variance

- Underfitting happens when a model unable to capture the underlying pattern of the data.
- It happens when we have very less amount of data to be build an accurate model.
- **When we try to build a linear model with a non-linear data.**
- The model is very simple and unable to capture the complex patterns in data.
- **Models are consistent but inaccurate on average**

+ High Bias – High Variance

- **This would have to be the worst situation.**
- **The error of prediction is very high.**
- The predictions also **fluctuate rapidly** with **change** in **training data**.
- Models are **inaccurate** and also **inconsistent** on average.

+ Identify the high variance & high Bias

- ***High Bias*** can be identified when we have
  - High training error
  - Validation error or test error is same as training error
- ***High Variance*** can be identified when
  - Low training error
  - High validation error or high-test error

How do we fix high bias or high variance in the data set?

- **High bias is due to a simple model,** and we also see a **high training error**.
- To fix that we can do following things
  - **Add more input features or add more importance to data**
  - **Add more complexity by introducing polynomial features**
  - **Decrease Regularization term**

- **High variance** is due to a model that **tries to fit most of the training dataset** points and hence **gets more complex**.
- To resolve high variance issues, we need to work on
  - **Getting more training data**
  - **Reduce input features**
  - **Increase Regularization term**

+ WHY Bias and Variance in ML?

- **Help to avoid over and underfitting conditions**
  - **To have consistency in predictions**

End of Lecture - 04